More Joy of Sets, $\boxed{\S}$14

Let S be a non-empty set.

Definition 1. *A set S is* **finite** *if there exists a bijection (one-to-one correspondence) $f : S \to \{1, 2, \ldots, n\}$ for some natural number n. The set S is* **infinite** *if no such bijection exists for any n.*

Remark: We also call the empty set finite.

Definition 2. *The* **size** *or* **cardinality** *of a finite set S is the unique n such that there exists a bijection from S to $\{1, 2, \ldots, n\}$. We denote the cardinality of S by $|S|$.*

Remark: We also define the cardinality of the empty set to be 0.

1. We've seen the formula

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

Find a formula for $|A \cup B \cup C|$.

2. Can you conjecture a more general formula for $|A_1 \cup A_2 \cup \cdots \cup A_k|$?

Definition 3. *Sets A and B* **have the same cardinality** *if there is a bijection from A to B.*

3. How do the sets $\mathbb{N}$ and $2\mathbb{N} = \{2n : n \in \mathbb{N}\}$ compare?

4. How do the sets $\mathbb{N}$ and $\mathbb{N}^2$ compare?

Definition 4. *An infinite set S is called* **countably infinite** *if there is a bijection from S to $\mathbb{N}$; otherwise S is* **uncountably infinite** *or* **uncountable***.*

5. Is the set $\mathbb{Z}$ countably infinite?

6. Compare the following sets.

(a) $A = (0, \infty)$ and $B = \mathbb{R}$.

(b) $A = (0, 1)$ and $B = \mathbb{R}$.

7. Compare $\mathbb{R}$ and $\mathbb{N}$.

Definition 5. *Define the* **power set** *of a set A as the set of all subsets of A; namely*

$$P(A) = \{X : X \subseteq A\}.$$

8. Let $A = \{1, 2\}$. Find the power set of A.

9. What is $|P(\{1, \ldots, n\}|$?

Theorem: There is no bijection from A to $P(A)$.

10. Prove the Theorem.

More on sets – Answers / Solutions

1. The idea of

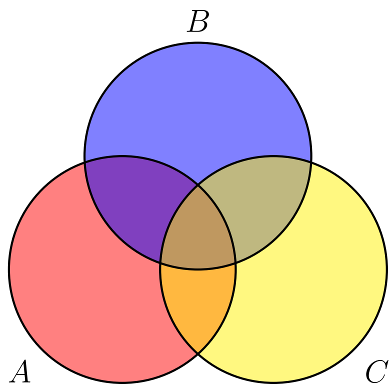
$$|A \cup B| = |A| + |B| - |A \cap B|$$

is that, in counting the elements of $A \cup B$, we double-count the elements that are in both A and B. That is, we double-count $A \cap B$.

With three sets the idea is similar, and we get

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|.$$

Here is a pictorial representation of this that could help you come to grips with this formula:

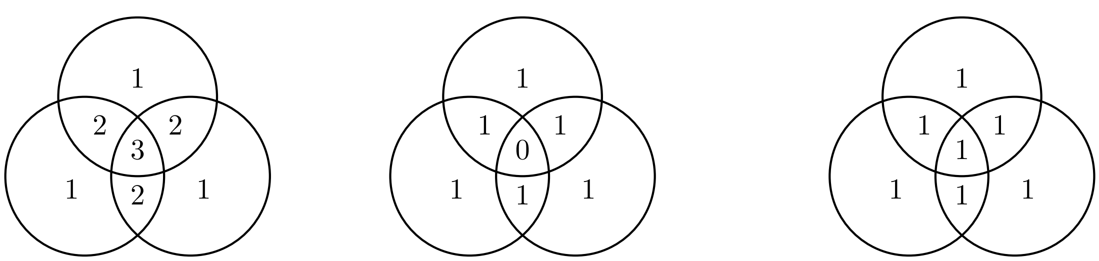


One way to think about the count is to check each piece. For example, if $x \in A \cap B$ but $x \notin C$, then x is counted positively in $|A|$ and $|B|$ and negatively in $|A \cap B|$. Thus x is counted a total of once. Elements in every region are similarly counted only once, as you can check.

2. The general expression is called the principle of inclusion-exclusion, and the general formula we're looking for can be formulated as

$$|A_1 \cup A_2 \cup \cdots \cup A_k| = \sum_{k=1}^{n} (-1)^{k+1} \left(\sum_{i_1 < i_2 < \cdots < i_k} |A_{i_1} \cap A_{i_2} \cap \cdots \cap A_{i_k}| \right).$$

Hopefully the name makes sense: we're including regions (but thus over-counting), then excluding regions (thus *under*-counting), and so on until the sum is correct. The Wikipedia article (linked above) has a nice diagram that shows the number of times each region is counted at each step in computing $|A \cup B \cup C|$. We try to reproduce it here:



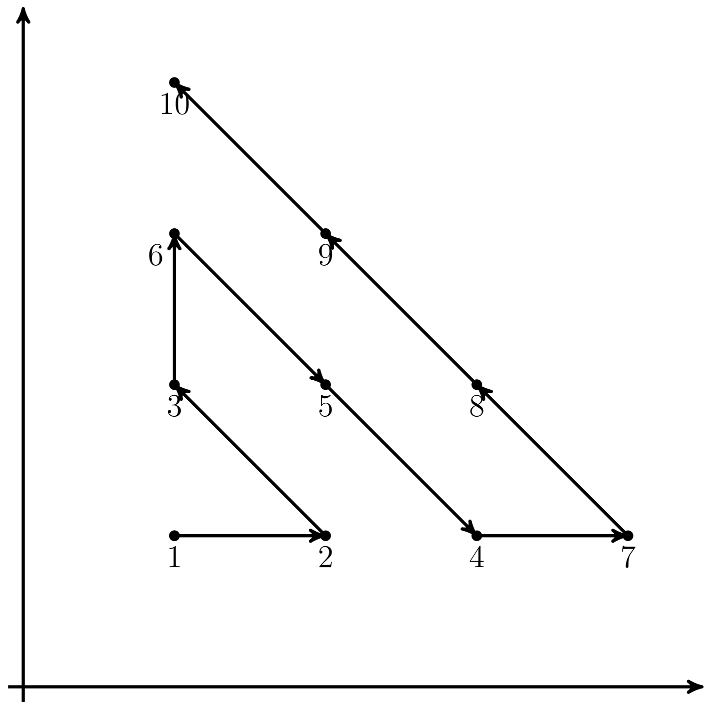
$|A| + |B| + |C|$

$|A| + |B| + |C|$
$-|A \cap B| - |A \cap C| - |B \cap C|$

$|A| + |B| + |C|$
$-|A \cap B| - |A \cap C| - |B \cap C|$
$+|A \cap B \cap C|$

3. Define $f : \mathbb{N} \to 2\mathbb{N}$ by $f(n) = 2n$, then f is a bijective. Thus the sets have the same cardinality, or $|\mathbb{N}| = |2\mathbb{N}|$.

4. First we give a graphical description of a bijection from $\mathbb{N}$ to $\mathbb{N}^2$. We enumerate the elements of $\mathbb{N}^2$ by starting at $(1, 1)$ and moving outwards along diagonals. Using this method each pair of natural numbers is assigned a unique natural number, and hence $|\mathbb{N}| = |\mathbb{N}^2|$.



Another approach is to define $f : \mathbb{N}^2 \to \mathbb{N}$ by $f(m, n) = 2^{m-1}(2n - 1)$.

Claim: Every natural number can be uniquely written as a product of an odd number and a power of 2.

Proof. If the claim fails, then there is some smallest n where the claim fails. Suppose n is the smallest natural number such that n doesn't have a unique expression as a product of an odd number and a power of 2. If n is odd, then n is not divisible by 2, so $1 \cdot n$ is such an expression and the only one. If n is even, then consider $n/2$. Each representation of n yields a corresponding representation of $n/2$, so $n/2$ is a smaller instance of the claim failing, which contradicts our assumption that n is the smallest. Thus there can be no n where the claim fails. $\square$

Using the claim, we get that f is a bijection.

5. Yes. Recall from the problem set $g : \mathbb{Z} \to \mathbb{N}$ by

$$g(k) = \begin{cases} 2k + 1 & k \geq 0, \\ -2k & k < 0. \end{cases}$$

We proved in the problem set that g is a bijection. Thus $|\mathbb{Z}| = |\mathbb{N}|$, and hence $\mathbb{Z}$ is countably infinite.

6. (a) Define $f : \mathbb{R} \to (0, \infty)$ by $f(x) = e^x$, then f is a bijection from $\mathbb{R}$ to $(0, \infty)$.

(b) Define $f : (0, 1) \to \mathbb{R}$ by $f(x) = \tan\left(\pi(x - \frac{1}{2})\right)$. We can check the function is a bijection.

7. Theorem: There is no bijection from $\mathbb{N}$ to $\mathbb{R}$.

Proof. We prove there is no biection from $\mathbb{N}$ to $(0, 1)$ by contradiction. Suppose $f : \mathbb{N} \to (0, 1)$ is a bijection. We express $f(n)$ as a decimal expansion (avoiding expansions ending in infinite 9's). Then

$$\begin{aligned} f(1) &= 0.a_{11}a_{12}a_{13}\ldots a_{1n}\ldots \\ f(2) &= 0.a_{21}a_{22}a_{23}\ldots a_{2n}\ldots \\ &\vdots \qquad\quad \ddots \\ f(n) &= 0.a_{n1}a_{n2}a_{n3}\ldots a_{nn}\ldots \\ &\vdots \end{aligned}$$

Now define $b \in (0, 1)$ by $b = 0.b_1 b_2 \ldots b_n \ldots$ where $b_k = \begin{cases} 1 & \text{if } a_{kk} = 2, \\ 2 & \text{if } a_{kk} \neq 2. \end{cases}$ Now $f(m) \neq b$ for any m since $f(m)$ and b differ in the m-th position of their decimal expansions. Thus f cannot be surjective, which contradicts our assumption that f was a bijection. $\square$

This proof technique is called Cantor diagonalization. We now have $|\mathbb{R}| > |\mathbb{N}|$, and $\mathbb{R}$ is uncountable.

8. Let $A = \{1, 2\}$, then $P(A) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$.

9. Claim: Let A be a finite set with n elements, then $|P(A)| = 2^n$.

Proof. We prove the claim by induction.

Base Case: When $n = 1$, we have $|P(A)| = |\{\emptyset, A\}| = 2^1$, and hence the base case holds.

Inductive Step: We assume if A is a set with n elements, then $|P(A)| = 2^n$. Let B be a set with $n+1$ elements; namely $B = \{b_1, \ldots, b_{n+1}\}$. Let $C = \{b_1, \ldots, b_n\}$, then C has n elements and hence there are 2^n possible subsets of C by the induction hypothesis. Every subset of C is also a subset of B. Any subset of B either contains b_{n+1} or not. If the subset does not contain b_{n+1}, then it is also a subset of C and there are 2^n possibilities. If the subset does contain b_{n+1}, then it is obtained from a subset C by including b_{n+1}. So there are 2^n possibile subsets containing b_{n+1}. Thus $|P(A)| = 2^n + 2^n = 2^{n+1}$. $\square$

10. Cantor's Theorem: There is no bijection from A to $P(A)$.

Proof. Suppose for a contradiction there is a bijection $f : A \to P(A)$. We want to show that f cannot be surjective by finding a subset of A that does not appear in the image of f. Consider $B = \{x \in A : x \notin f(x)\}$, then $B \subseteq A$ and hence $B \in P(A)$. Since f is bijective, there exists $a \in A$ such that $f(a) = B$. Now there are two possibilities $a \in B$ or $a \notin B$.

Case 1: Suppose $a \in B$, then by definition of B, we know that $a \notin f(a)$, which means that $a \notin B$. Thus this case is impossible

Case 2: Suppose $a \notin B$, then $a \notin f(a)$. Thus $a \in B$ from the definition of B. Thus this case is also impossible.

Since both cases are impossible, our assumption that f is bijective must be impossible. $\square$